

Xiangling Xu (许湘灵)

My Website

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G Google Scholar

ID Orcid



Research Interests: Quantum information, quantum foundations, operator algebras, noncommutative polynomials optimization

Education

- 2023 – now **Ph.D., Inria Saclay Île-de-France & École doctorale IP Paris**
Supervisor: Marc-Olivier Renou
Project: *Quantum correlations in causal structures: characterization and application*
- 2020 – 2023 **M.Sc. Mathematics, ETH Zürich**
Master Thesis: *Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective*
Semester Project: *Quantum BCOV Theory on Calabi-Yau Manifold*
CGPA: 5.69/6.00
- 2016 – 2020 **B.Sc. Mathematics and Physics, University of Toronto**
Projects: *Noncommutative Geometry in Gravity, Large-Scale Cosmological Structures, Optical Simulation, Biomechanical Simulation*
CGPA: 3.95/4.00

Publications and Preprints

- 1 M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, and **X. Xu**, *Quantitative quantum soundness for all multipartite compiled nonlocal games*, 2025. arXiv: 2509.25145 [quant-ph]. [URL: https://arxiv.org/abs/2509.25145](https://arxiv.org/abs/2509.25145).
- 2 I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, **X. Xu**, and Y. Zhao, *Quantitative quantum soundness for bipartite compiled bell games via the sequential npa hierarchy*, 2025. arXiv: 2507.17006 [quant-ph]. [URL: https://arxiv.org/abs/2507.17006](https://arxiv.org/abs/2507.17006).
- 3 **X. Xu**, M.-O. Renou, and I. Klep, *Quantitative tsirelson's theorems via approximate schur's lemma and probabilistic stampfli's theorems*, 2025. arXiv: 2505.22309 [quant-ph]. [URL: https://arxiv.org/abs/2505.22309](https://arxiv.org/abs/2505.22309).
- 4 M.-O. Renou, **X. Xu**, and L. T. Ligthart, *Two convergent npa-like hierarchies for the quantum bilocal scenario*, 2024. arXiv: 2210.09065 [quant-ph]. [URL: https://arxiv.org/abs/2210.09065](https://arxiv.org/abs/2210.09065).
- 5 X. Luo, **X. Xu**, and X. Wang, "On the kinematic morphology around haloes," *Monthly Notices of the Royal Astronomical Society*, vol. 518, no. 4, pp. 6059–6064, 2023. [DOI: https://doi.org/10.1093/mnras/stac3500](https://doi.org/10.1093/mnras/stac3500).
- 6 **X. Xu**, "Quantum nonlocality in bilocal networks: An operator algebraic perspective," M.S. thesis, ETH Zurich, 2023. [DOI: https://doi.org/10.3929/ethz-b-000613670](https://doi.org/10.3929/ethz-b-000613670).

Conferences, Seminars, Workshops

Invited Talks

Jun 2026 **SIOPT 2026 Edinburgh**, University of Edinburgh

Conferences, Seminars, Workshops (continued)

Jul 2025  *IWOTA 2025 Twente*, University of Twente

Contributed Talks

Oct 2025  *YQIS25*, Institute of Photonic Sciences (ICFO)

Jun 2025  *IQC-PCQT-Quantum Saclay Workshop*, Sorbonne University

Sep 2024  *Causalworlds*, Perimeter Institute for Theoretical Physics

Nov 2023  *Workshop Defi EQIP*, Inria Lyon

Oct 2023  *Saclay Quantum Seminar*, Inria Saclay

Poster Sessions

Nov 2024  *Colloquium GDR TeQ*, Sorbonne University

 *YQIS24*, Inria de Paris

Other Participation

Mar 2025  828. WE-Heraeus-Seminar: Operator Theory and Polynomial Optimization in Quantum Information Theory, Bad Honnef

Sep 2024  VQF Vienna Quantum Foundations Conference, IQOQI Vienna

Nov 2023  POP23 Workshop on Future Trends in Polynomial OPTimization, LAAS-CNRS Toulouse

Collaborations

Group Visits

Oct-Nov 2025  *Thomas Vidick*, EPFL, Lausanne
Invited two-week visit

Sep 2024  *Igor Klep*, University of Ljubljana
Invited two-week visit funded by Erasmus+ Mobility program

Jul 2024  *William Slofstra*, Institute for Quantum Computing, University of Waterloo
Invited two-week visit

Dec 2023  *David Gross*, University of Cologne
Invited one-week visit

Organizations

Mar 2025 - Now  Coorganizer for *Seminars in Commutative and Noncommutative Polynomial Optimization for Quantum Theory*
Coordinating the Gaspard Monge Visiting Professor program for Igor Klep
Co-planning the seminar and the schedule for invited speakers

Mar 2025  Coorganizer for *828. WE-Heraeus-Seminar at Bad Honnef*
Exploring Operator Theory and Polynomial Optimization in Quantum Information Theory with world-renowned experts
Coordinating with the organizers and inviting some of the speakers

Feb 2024 - Now  Coorganizer of *Project COMPUTE for Quanterra*
Coordinating with the members
Making the project poster and website

Collaborations (continued)

- Sep 2023  Coorganizer for *Saclay Quantum Seminar on NPA hierarchy*
Planning and organizing the one-day program to bring all quantum groups at Saclay together

Awards and Scholarships

- 2024  *Erasmus+ Mobility Program*, funds for visiting the University of Ljubljana
- 2020  *McNab Undergraduate Scholarship*, University of Toronto
- 2017 - 2020  *Dean's List Scholar*, University of Toronto
- 2018  *The Class of 3To and Associates Scholarship in Mathematics and Physics*, University of Toronto
-  *International Exchange Award*, University of Toronto
-  *Bronze Medal*, 2017 University Physics Competition
- 2017  *Eileen Doman Award*, University of Toronto
-  *Summer Undergraduate Research Program*, Canadian Institute for Theoretical Astrophysics
- 2016  *President's Scholar of Excellence Program*, University of Toronto

Teaching Experience

Lecture

- Nov 2024  *Recent trends in parallel, distributed, and quantum computing*
Master 2 QDCS, Université Paris-Saclay

Teaching Assistance

- 2022-2023  *Linear Algebra I & II*
Department of Mathematics, ETH Zürich
- 2021-2022  *Analysis I & II*
Department of Mathematics, ETH Zürich

Skills

- Languages  Chinese (native), English (fluent), German (beginner)
- Coding  Python (Ncpol2sdpa), Matlab (Yalmip), Mathematica, C Language, Hugo on GitHub, $\LaTeX$

Referees

Available on Request